ECONOMICS OF CROP SYSTEM IN KARNATAKA- A CASE STUDY OF KOLAR DISTRICT

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Abstract

This paper attempted to study to analyze the cropping patterns in Karnataka and also it focuses on analysing the various crops produced in the Study area. To achieve these goals the researcher has collected the necessary information from various data sources such as the 2011 census survey papers, journals, newspapers, magazines, websites, etc. the same information is analysed systematically. After analysis, it concludes that India is an agricultural economy and the agricultural sector employs about half of the country's workforce and contributes about 16 percent of GDP during the 2013-2013 financial year. This study found that the landholdings for the year 2005-06 of different sizes are given in the above table. From the table, it is observed that the number of marginal holdings below one hectare and smallholdings of size one to two hectares constitute 84.46 percent of the total holdings. Only 0.46 percent of holders own land of more than ten hectares. Niger seed is covered 223 hectares with a production of 78 tonnes. And yield in KGs per hector is 354. Similarly Rape and Mustard covered the 285 area in hectares with a production of 77 tonnes per year with a yield in kgs per heart 413. Sunflower is coveted 51 hectors with 20 tonnes production in the year but it shows that yield in KGs per hector is 413 which is highest among the Oil Seeds. The majority of the area is covered with the Groundnut with 7878 area of hector with the production of 7052 tonnes and yield per kgs per hector 942.

Keywords: Cropping, Agricultural land, Irrigation, Production, Yield per hector

1. INTRODUCTION

Agriculture still forms the backbone of the Indian economy, despite concerted efforts towards industrialization in the last decades. Agriculture contributes a high share of domestic products by sectors in India. Farmers are growing various crops in the field rather than a single crop. The physical factors determine the outline of the area of crops, while the socio-economic relationships determine their extent and government policy can impact crop growing like new various crops can be introduced in the place of traditional crops. The general land use pattern of the state has to be seen in the light of its overall natural and changing socio-cultural conditions. Its land-use zones include cultivated land, barren and uncultivated land, land under forest and pasture, etc. Human and economic factors have influenced the changes in the cropping pattern of Karnataka. The southern part of the state account for 42.22% of the state's agricultural land of which only 14.00% is irrigated. However, the remaining 57.78 percent lies in the northern parts. Irrigational facilities are available for 20% only. Nearly 11.82 percent of the total area is not available for cultivation baring either or put to non-agricultural uses and 8.58 falls under other cultivated lands including the permanent grasslands, cultivable waste, and miscellaneous tree crops and groves. The fallow land is 9.27 and the net has sown 54.69% of which is the area is sown more than once i.e. 12.98 percent added.

The northern regions of the state reflect marked pattern contrasts in their spatial crop distributions and thus when the northeastern part dominates in the raising of Jower, Bajra, Miza, wheat, and pulses and the northwestern part of the state dominate the sugarcane and cotton cultivation. The southern parts of the state reflect the noticeable patterns of crops distribution. It dominates the paddy, Ragi (staple crop of southern Karnataka), and oilseeds. The state strongly follows the rice cutting pattern. Different crops instead of rice, rag, bajra, cotton, nuts, jowar, and maize.2 Other important crops are wheat and small grains and pulses such as tur, Bengal gram, horse gram, black gram, green gram, cowpea, etc. Oilseeds include nuts. , sesame, sunflower, soybeans, and sunflowers. Commercial crops include sugarcane in the eastern region, cotton in the northwestern region, and tobacco. Cashew, coconut, areca nut (southern region), cardamom, and peppers are other important plants. The Western Ghats are well-known for their coffee and tea plantations while maize is widely grown in the northern region of the region. Due to the climate, the coastal region is well suited for the cultivation of orchards. The paddy field, maize, pulses, sugarcane, and tobacco have recorded amazing growth over the past 50 years. While paddy grew from 10.28 lakh ha in 1961 to 15.1 lakh ha in 2009, maize grew from 11,000 hectares in 1961 to 12.9 lakh ha in 2009. lakh ha from 1961 to 2001. Over the past 50 years, the planting area has seen a slight decrease from 102.3 lakh ha (1961) to 101.7 lakh ha (2009) while the planting area increased by about 17% from 105.9 lakh ha to 123.7 lakh ha at the same time.3 Horticulture covers 18.9 lakh ha with a production of 136.6 lakh ha. Fruits (mango, banana, papaya, grapes, sap, etc.) account for 41.9%; vegetables (potatoes, tomatoes, onions, brinjal, etc.) 45.3%; spices

2. AGRICULTURAL LAND USE / CROPPING PATTERN OF THE KARNATAKA

Agricultural land use means land under net sown area, fallow land, and uncultivable land excluding fallow land. In short agricultural land use means a cropping pattern. It means the proportion of area under various crops at a point of time/ yearly sequence and spatial arrangements of crops and fallow on a given area. A cropping pattern is a dynamic concept as it changes over space and time. It is influenced by geographical, social, economical, and political factors. The net sown area in Karnataka is normally about 11.2 million hectares. Only 24 percent of the arable area is under irrigation, so most of the cultivable area depends on the occurrence and distribution of pre-monsoon and southwest monsoon precipitation. The cropping pattern of Karnataka is dominated by crops planted during the Kharif southwest monsoon season (June to October). Normally, 65 percent of the net sown area is planted with crops during Kharif depending on pre-monsoon and southwest monsoon rains. Similarly, about 30 percent and 5 percent of the area is sown during Rabi, the northeast monsoon season (November to

March), and the summer season (April to June), respectively, depending on the quantum of residual moisture. The agricultural land in Karnataka is 65.83 percent of the total geographical area. This includes 54.69 percent area of the net sown area and 12.98 percent area is shown more than once. The cropping pattern of the state is typical that the food crops cover most of the cultivated area. In the state, food crops occupy the largest area which is 75.55 percent of the total net sown area in 2008-09. Which paddy is the leading crop followed by Jower, Maize, and other food grains that occupy a small proportion of the area? Paddy is the main Kharif crop. Which is grown from June to October thought the state; it is also as grown as a rabbi season with the help of irrigation facilities in a few districts of the state. Jower, Bajra are other main Kharif crops whereas Ragi and small millets come next and are grown on a limited area. (Ragi is a major crop in south Karnataka than the other food crops) among non-food/commercial crops sugarcane is the dominant which is 8.72 percent of the net area sown of the state.

Rainfall Pattern in Karnataka

Two-thirds of Karnataka's geographical area is arid or semi-arid. Out of 30 districts, 19 districts are drought-prone with annual normal rainfall of less than 750 mm. The normal annual rainfall of the state is 1,139 mm, received over 55 rainy days. Of the annual rainfall, 71 percent is received during the Kharif season, 17 percent during the Rabi season, and the remaining 12 percent during the pre-monsoon season. Because rainfall is highly variable over space and time and irrigation is limited, agricultural production is correspondingly variable. Even during the good rainfall years, at least 25 percent of the taluks [district subdivisions] in the state are affected by the uneven distribution of rainfall, and even the assured rainfall areas like the coastal region can experience drought-like conditions.

3. PROFILE OF STUDY AREA

Kolar is the district headquarters. Located in southern Karnataka, it is the state's easternmost district. The district is surrounded by the Bangalore Rural district on the west, Chikballapur district on the north, the Chittoor district of Andhra Pradesh on the east, and the Krishnagiri district of Tamil Nadu on the south. On 10 September 2007, it was bifurcated to form the new district of Chikballapur. Due to the discovery of the Kolar Gold Fields, the district has become known as the "Golden Land" of India. People are citing that still, gold is present in Kolar Gold Fields mines abundantly and also exists in Mulbagal, Kolar, Bangarapet, Malur, Srinivasapura taluks of Kolar District. However, it must have been confirmed by the state and as well central govt authorities.

(1) Cropping pattern

As per the 2011 Census, there are 25.50 percent cultivators and 28.03 percent agricultural laborers. Paddy, Ragi, and maize are the major crops grown in the district. Groundnut, sunflower, Niger seed, and rape & mustard are the other oil seeds crop grown in the district. Among the commercial crops, mention may be made of sugar cane only. A sufficient and considerable amount of land is put under plantation and horticultural crops. The chief crops grown are potato, cabbage, banana, onion, tomato, cashew nut, grapes, etc. The other condiments and spices produced in the district are dry chilies and coriander. The yield of paddy under irrigated land per hectare is 3,237 kg. It is 2,055 kg. per hectares under unirrigated conditions. The maize per hectare is 2,689 kg. under irrigated conditions whereas it is 2,418 kg. per hectare under unirrigated condition.

(2) Irrigation

Scanty rainfall and the uncertainty of a well-distributed rainfall highlight the importance of irrigation. There are no perennial rivers or big water storage. Hence, irrigation by tanks and wells is a characteristic feature of the district. In the matter of exploitation of underground water resources, Kolar district has earned an important place among the districts in the State. Tanks and wells are the chief sources of irrigation in the district. The district has offered immense potentialities for the construction of tanks because of its peculiar river system. These advantageous features have already been adequately exploited as well for it is evident that in the Kolar district the tank system is very well established. Most of the tanks are found in series are emptying into the other. A large number of tanks is constructed on a connected system of streams and their feeders which are many. Most of the wells in the district are fitted with electric pump sets. In addition, there are also yathas and piccolos which are operated manually. With no major or perennial river coursing through the district, there is no scope for any major irrigation project.

4. SIGNIFICANCE OF THE STUDY

This study helps to former to decide on their crops. This study helps Marketers to set the appropriate strategy for their business. This study helps the government to chalk out the plans and policy towards the various Agricultural schemes towards crops in the study area. This study will be input for future Academicians, Scholars, and Researchers. This study helps various stakeholders of the crops who are directly or indirectly getting the benefits from crops

5. OBJECTIVES OF THE STUDY

- To analyses the cropping patterns in the Karnataka
- To analyze the various crops produced in the Kolar district.

6. RESEARCH METHODOLOGY

A research design is the arrangement of conditions for the collection and analysis of data in a manner that aims to combine relevant research purposes with the economy in procedure. The research design is analytical and descriptive.

S.No	Particulars	Research Design
1.	Research Design	Analytical Research Design
2.	Data Collection	Secondary data
3.	Source of Data Collection	2011 census Reports of Kolar district, Journals, Articles, RBI, Websites, Newspaper etc.

7. DATA ANALYSIS AND INTERPRETATION

TABLE NO 1 LAND HOLDINGS IN HECTARES 2005-2006

Name of Taluk	Marginal (Below 1)		Small (1–2)		Semi-Medium (2-4)		Medium (4-10)		Large (10 & above)		Total Holdings	
	Holder	Area	Holder	Area	Holder	Area	Holder	Area	Holder	Area	Holder	Area
Srinivaspur	22820	10183	7472	10567	3499	9430	1501	8769	235	3380	35527	42329
Kolar	29602	13265	9810	13698	4640	12510	1469	8226	119	1672	45640	49371
Malur	28049	12225	9084	12489	4319	11602	1472	8276	123	1644	43047	46236
Bangarapet	24294	12013	10672	14524	6232	16277	2596	13892	376	4653	44170	61366
Mulbagal	27741	12597	10702	14793	4822	12933	1619	9138	140	1907	45024	51368
Total	**132506**	**60290**	**47740**	**66071**	**23512**	**62752**	**8657**	**48301**	**993**	**13256**	**213408**	**250670**

(Source: Fully revised estimates of principal crops in Karnataka for the year 2009-2010, Directorate of Economics and Statistics, Bangalore.)

INTERPRETATION

The Above Table No 1 Shows the land holdings in hectares 2005-2006 The size of the agricultural holdings is one of the important factors that determine the productivity of the land. The landholdings for the year 2005-06 of different sizes are given in the above table. From the table, it is observed that the number of marginal holdings below one hectare and smallholdings of size one to two hectares constitute 84.46 percent of the total holdings. Only 0.46 percent of holders own land of more than ten hectares.

TABLE NO 2 AREAS, PRODUCTION AND AVERAGE YIELD OF CROPS FOR 2009-10 UNDER ALL SEASONS

OILSEEDS			
Crops	**Area in hectares**	**Production intonnes**	**Yield in Kgs per hectare**
Sesamum	0	0	0
Linseed	0	0	0
Nigerseed	223	78	354
Rape & Mustard	285	77	273
Sunflower	51	20	413
Soyabean	0	0	0
Safflower	0	0	0
Groundnut	7878	7052	942
Crop Total Oil seeds	8509	7274	900

(Source: Fully revised estimates of principal crops in Karnataka for the year 2009-2010, Directorate of Economics and Statistics, Bangalore.)

INTERPRETATION

The above table No 2 shows the area, production, and average yield of crops for 2009-10 under all seasons. It shows that Niger seed is covered 223 hectares with a production of 78 tonnes. And yield in KGs per hector is 354. Similarly Rape and Mustard covered the 285 area in hectares with a production of 77 tonnes per year with a yield in kgs per heact 413. Sunflower is coveted 51 hectors with 20 tonnes production in the year but it shows that yield in KGs per hector is 413 which is highest among the Oil Seeds. The majority of the area is covered with the Groundnut with 7878 area of hector with the production of 7052 tonnes and yield per kgs per hector 942.

TABLE NO 3 AREAS, PRODUCTION AND AVERAGE YIELD OF CROPS FOR 2009-10 UNDER ALL SEASONS

COMMERCIAL CROPS			
Crops	**Area in hectares**	**Production in tonnes**	**Yield in Kgs perhectare**
Cotton	0	0	0
Sugarcane	300	0	0
Tobacco	0	0	0
Sunhemp	0	0	0
Mesta	0	0	0

(Source: Fully revised estimates of principal crops in Karnataka for the year 2009-2010, Directorate of Economics and Statistics, Bangalore.)

INTERPRETATION

The above table 3 above shows the area, production, and average yield of Commercial crops for 2009-10 under all seasons. Among the 5 commercial crops, sugarcane is covered 300 hectares in area.

TABLE NO 4 AREA, PRODUCTION AND AVERAGE YIELD OF CROPS FOR 2009-10 UNDER ALL SEASONS

PLANTATION AND HORTICULTURAL CROPS			
Crops	**Area in hectares**	**Production in Tonnes**	**Yield in Kgs perhectare**
Coconut	1471	10459	7182
Onion	292	1525	5497
Castor	72	47	680
Tomato	7589	71919	9477
Brinjal	299	2066	6910
Banana	568	8145	14340
Grapes	105	1861	17720
Mango	33131	136135	4109
Papaya	131	4370	33699
Cashewnut (rawnuts)	883	670	766
Guava	430	417	969
Sapota	1583	2446	1545

Cashew nut (processed)	883	168	192
Lemon	22	178	8076
Sweet Potato	0	0	0
Potato	6154	89663	15337
Beans	1555	11072	7120
Cabbage	370	6013	16252
Tapioca	0	0	0
Pomogranate	0	0	0
Arecanut (Raw)	0	0	0
Arecanut (Processed)	0	0	0

(Source: Fully revised estimates of principal crops in Karnataka for the year 2009-2010, Directorate of Economics and Statistics, Bangalore.)

INTERPRETATION

The above table no 4 shows the area, production, and average yield of plantation and horticultural crops for 2009-10. Through the above table, it’s found that the Mango crops are covered by the 33131 hectors with the production of 136135 tonnes, yield in KGs per hector shows the 4109, followed d by this tomato crop is covered the 7589 hectors with the production of 71919 tonnes. And yield in KGs per hector shows 9477 tonnes.

TABLE NO 5 AREAS, PRODUCTION AND AVERAGE YIELD OF CROPS FOR 2009-10 UNDER ALL SEASONS

CONDIMENTS AND SPICES CROPS			
Crops	**Area in hectares**	**Production in tonnes**	**Yield in Kgs perhectare**
Dry Chillies	1589	2960	1961
Turmeric	0	0	0
Garlic	0	0	0
Coriander	5	2	495
Dry Ginger	0	0	0
Black Pepper	0	0	0
Cardamom	0	0	0

(Source: Fully revised estimates of principal crops in Karnataka for the year 2009-2010, Directorate of Economics and Statistics, Bangalore.)

INTERPRETATION

The Above table no 5 shows the area, production, and average Yield of Condiments and Spices Crops for 2009-10. Through the above table, it’s found that the dry chilies crops are covered the 1589 hector with the production of 2960 tonnes, yield in KGs per hector shows 1961.

8. CONCLUSION

The agriculture sector of Karnataka has undergone wide-ranging changes in terms of ownership of land, cropping pattern, cultivation practices, productivity, and intensity of cultivation. Unlike the other regions in India, the farm front of Karnataka is characterized by extreme diversity in its bio-physical resource base and agro-climatic endowments providing multiple opportunities for raising a variety of crops. It has a wide range of cropping patterns that vary widely from region to region and to a lesser extent from one year to another year, there are various ways of utilizing the land intensively. This study found that the landholdings for the year 2005-06 of different sizes are given in the above table. From the table, it is observed that the number of marginal holdings below one hectare and smallholdings of size one to two hectares constitute 84.46 percent of the total holdings. Only 0.46 percent of holders own land of more than ten hectares. Niger seed is covered 223 hectares with a production of 78 tonnes. And yield in KGs per hector is 354. Similarly Rape and Mustard covered the 285 area in hectares with a production of 77 tonnes per year with a yield in kgs per heact 413. Sunflower is coveted 51 hectors with 20 tonnes production in the year but it shows that yield in KGs per hector is 413 which is highest among the Oil Seeds. The majority of the area is covered with the Groundnut with 7878 area of hector with the production of 7052 tonnes and yield per kgs per hector 942

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